

A 15-VERTEX GRAPH OF MINIMUM DEGREE TWO WITH $\alpha_{\text{bnr}} > \Gamma_b$

ABSTRACT. For a graph G let $\Gamma_b(G)$ be its upper broadcast domination number and $\alpha_{\text{bnr}}(G)$ its irredundant boundary independence broadcast number. In the last section of their paper on the two parameters, Mynhardt and Neilson ask two questions: whether $\alpha_{\text{bnr}}(G) \leq \Gamma_b(G)$ when G is 2-connected, and whether the same holds when $\delta(G) \geq 2$. We exhibit a graph G' on 15 vertices and 26 edges with $\delta(G') = 2$ and

$$\alpha_{\text{bnr}}(G') \geq 6 > 5 = \Gamma_b(G'),$$

The printed second sentence is malformed, and we answer the repair that reads it as a question about arbitrary G with $\delta(G) \geq 2$: under that reading the answer is *no*. The tool is a doubling lemma: if p is an end-vertex of a connected graph H and p' is a true twin of p , then $\Gamma_b(H + p') = \Gamma_b(H)$. Here G' is the graph G_1 of their Section 4.2 with each of its four end-vertices doubled into a true twin, so the upper bound is inherited from their own published value $\Gamma_b(G_1) = 5$; the lower bound is a single exhibited broadcast. A second repair of the printed sentence is possible — whether $\alpha_{\text{bnr}} - \Gamma_b$ can be arbitrarily large when $\delta(G) \geq 2$ — and is not settled here: no member of the family G_k with $k \geq 2$ is treated. The *first* question, for 2-connected G , is not addressed either; G' has four cut vertices.

1. THE QUESTION

Mynhardt and Neilson [2] close their paper with four open questions. The fourth reads, verbatim,

Is it true that $\alpha_{\text{bnr}}(G) \leq \Gamma_b(G)$ when G is 2-connected? Is it true that $\alpha_{\text{bnr}}(G_k) \leq \Gamma_b(G_k)$ when $\delta(G) \geq 2$?

That is *two* questions, and only the second is at issue here:

Question (the second sentence of [2, Question 4]). Is it true that $\alpha_{\text{bnr}}(G) \leq \Gamma_b(G)$ when $\delta(G) \geq 2$?

The printed second sentence carries the subscript G_k under a hypothesis naming an unbound G , and $\delta(G_k) = 1$ in that same paper, so it is answered here in the repaired reading displayed above, which is the evident intended one. The paragraph preceding it states the grounds for asking: “We showed in Theorem 4.9 that $\alpha_{\text{bn}} - \Gamma_b$ and $\alpha_{\text{bnr}} - \Gamma_b$ can be arbitrary for cyclic graphs, but the graphs used in the proof have end-vertices.” That sentence admits

2020 *Mathematics Subject Classification*. 05C69, 05C12.

Key words and phrases. broadcast domination, upper broadcast domination number, boundary independent broadcast, private boundary, true twin, counterexample.

a second repair, namely whether $\alpha_{\text{bnr}} - \Gamma_b$ can be arbitrarily large when $\delta(G) \geq 2$ — the natural continuation of the quoted Theorem 4.9 remark. This note does not settle that reading: it exhibits one graph with $\delta = 2$ and $\alpha_{\text{bnr}} > \Gamma_b$, and treats no member of G_k with $k \geq 2$.

Definitions, as in [2]. A *broadcast* on a connected graph G is a function $f : V(G) \rightarrow \{0, 1, \dots, \text{diam } G\}$ with $f(v) \leq e(v)$ for every v , where $e(v)$ is the eccentricity of v ; its *weight* is $\sigma(f) = \sum_v f(v)$ and $V_f^+ = \{v : f(v) > 0\}$ is its set of *broadcasters*. Write $N_f(v)$ for the ball and $B_f(v)$ for the sphere of radius $f(v)$ about v . For $v \in V_f^+$ the *private boundary* is

$$\text{PB}_f(v) = \left(\Omega_f(v) \setminus \bigcup_{z \in V_f^+, z \neq v} N_f(z) \right), \quad \Omega_f(v) = \begin{cases} N_f(v), & f(v) = 1, \\ B_f(v), & f(v) \geq 2; \end{cases}$$

so a broadcaster of strength 1 belongs to its own private boundary and one of strength at least 2 does not. Then f is *irredundant* if $\text{PB}_f(v) \neq \emptyset$ for every $v \in V_f^+$, and *bn-independent* if $N_f(u) \cap N_f(v) \subseteq B_f(u) \cap B_f(v)$ for all distinct $u, v \in V_f^+$; *no domination is required in either notion*. Finally $\alpha_{\text{bn}}(G)$, $\alpha_{\text{bnr}}(G)$ are the largest weights of a bn-independent, respectively bn-independent and irredundant, broadcast, and $\Gamma_b(G)$ is the largest weight of a *minimal dominating* broadcast, equivalently (the characterisation used throughout [2], going back to Erwin [1]) of a dominating irredundant broadcast. The asymmetry used below is already visible: a *non-dominating* broadcast is admissible for α_{bnr} and inadmissible for Γ_b . The phenomenon $\alpha_{\text{bnr}} > \Gamma_b$ is credited in [2] to [3].

2. THE GRAPH

Let G' be the graph on the 15 vertices

$$u_1, u_2, w_1, w_2, x_1, x_2, y_1, y_2, z_1, z_2, v, w'_1, w'_2, x'_1, x'_2$$

(in that order, indexed $0, \dots, 14$) with the following 26 edges:

$u_1 u_2, u_1 v, u_2 v$	$\{v, u_1, u_2\}$ a triangle, v has no other neighbour
$z_1 z_2, u_1 z_1, u_1 z_2, u_2 z_1, u_2 z_2$	$\{u_1, u_2, z_1, z_2\}$ a K_4
$y_1 z_1, y_1 z_2, y_2 z_1, y_2 z_2$	each y_i joined to both z_j
$u_1 y_1, u_2 y_2$	
$x_1 y_1, x_2 y_2$	
$x'_1 x_1, x'_1 y_1, x'_2 x_2, x'_2 y_2$	$\{x_i, x'_i, y_i\}$ a triangle, $N[x'_i] = N[x_i]$
$w_1 z_1, w_2 z_2$	
$w'_1 w_1, w'_1 z_1, w'_2 w_2, w'_2 z_2$	$\{w_i, w'_i, z_i\}$ a triangle, $N[w'_i] = N[w_i]$

Equivalently, G' is the labelled graph with graph6 string

`N_?CQJbs}?CCCCA_?g?`

on the vertex order above. Its degrees and eccentricities are

t	u_1	u_2	w_1	w_2	x_1	x_2	y_1	y_2	z_1	z_2	v	w'_1	w'_2	x'_1	x'_2
$\deg t$	5	5	2	2	2	2	5	5	7	7	2	2	2	2	2
$e(t)$	3	3	3	3	4	4	3	3	2	2	3	3	3	4	4

so $\delta(G') = 2$, $\text{diam } G' = 4$, and the broadcast space of G' has $\prod_t (e(t) + 1) = 1,474,560,000$ elements. The cut vertices of G' are exactly y_1, y_2, z_1, z_2 ; in particular G' is *not* 2-connected, so it is admissible for the question of Section 1 and inadmissible for the first sentence of [2, Question 4].

Deleting x'_1, x'_2, w'_1, w'_2 leaves a graph on 11 vertices and 18 edges whose end-vertices are exactly x_1, x_2, w_1, w_2 ; this is the graph G_1 of [2, Section 4.2], the member $k = 1$ of their family. Thus G' is G_1 with each of its four end-vertices doubled into a true twin. The identification is what makes the upper bound quotable; the program of Section 4 returns for that subgraph the published values $\Gamma_b(G_1) = 5$ and $\alpha_{\text{bnr}}(G_1) = 6$ of [2, Propositions 4.7 and 4.8] ($\Gamma_b(G_k) = 2k + 3$ at $k = 1$).

3. THE COUNTEREXAMPLE

Theorem 1. G' is a connected graph on 15 vertices with $\delta(G') = 2$ and

$$\alpha_{\text{bnr}}(G') \geq 6 > 5 = \Gamma_b(G').$$

Consequently the answer to the question of Section 1 is no.

Both bounds are by hand. The exact value $\alpha_{\text{bnr}}(G') = 6$ is not needed for Theorem 1 and is not asserted by it; it comes from the census described in Section 4.

Proof of $\alpha_{\text{bnr}}(G') \geq 6$, by hand. Take

$$f(x_1) = f(x_2) = 2, \quad f(w_1) = f(w_2) = 1, \quad f \equiv 0 \text{ elsewhere},$$

so $\sigma(f) = 6$; this is a legal broadcast because $e(x_i) = 4$ and $e(w_i) = 3$. Reading the balls and spheres off the edge list,

$$\begin{aligned} N_f(x_i) &= \{x_i, x'_i, y_i, u_i, z_1, z_2\}, & B_f(x_i) &= \{u_i, z_1, z_2\}, & \text{PB}_f(x_i) &= \{u_i\}, \\ N_f(w_i) &= \{w_i, w'_i, z_i\}, & B_f(w_i) &= \{w'_i, z_i\}, & \text{PB}_f(w_i) &= \{w_i, w'_i\}, \end{aligned}$$

for $i = 1, 2$; the broadcasters w_i lie in their own private boundaries because $f(w_i) = 1$, the broadcasters x_i do not because $f(x_i) = 2$. All four private boundaries are non-empty, so f is irredundant. For bn-independence it is enough that $d(u, v) \geq f(u) + f(v)$ for the six pairs of broadcasters, and indeed $d(x_1, x_2) = 4 \geq 2 + 2$, $d(x_i, w_j) = 3 \geq 2 + 1$ and $d(w_1, w_2) = 3 \geq 1 + 1$. Hence f is a bn-independent irredundant broadcast of weight 6 and $\alpha_{\text{bnr}}(G') \geq 6$.

Exactly one vertex is unheard by f , namely v , at distance 3 from every broadcaster. That is not an accident but the whole mechanism: a bn-independent irredundant *dominating* broadcast of weight 6 would be a minimal dominating broadcast of weight 6 and would contradict $\Gamma_b(G') = 5$. $\square$

The upper bound rests on the following lemma. Recall that p is an *end-vertex* of H if $\deg_H p = 1$, and that p' is a *true twin* of p in $H + p'$ if $N[p'] = N[p]$ there; for an end-vertex p with neighbour q this means exactly that p' is joined to p and to q , so that $\{p, p', q\}$ is a triangle and $\deg p = \deg p' = 2$.

Lemma 2 (end-vertex doubling). *Let H be a connected graph, p an end-vertex of H with neighbour q , and $H' = H + p'$ where p' is a true twin of p . Then*

$$\Gamma_b(H') = \Gamma_b(H).$$

Proof. Throughout, d denotes distance in H' . First, H' changes nothing metric about H . For $a, b \in V(H)$ we have $d(a, b) = d_H(a, b)$, since a path through p' can be shortcut through q . For $z \notin \{p, p'\}$ every $p'z$ -path and every pz -path leaves through q , whence

$$(1) \quad d(p', z) = 1 + d(q, z) = d(p, z) \quad (z \notin \{p, p'\}), \quad d(p, p') = 1.$$

Consequently $e_{H'}(a) = \max(e_H(a), d(a, p')) = e_H(a)$ for every $a \in V(H)$ and $e_{H'}(p') = e_H(p)$: eccentricities, and therefore the constraints $f(t) \leq e(t)$ defining a broadcast, are unchanged. Note also that the transposition of p and p' is an automorphism of H' .

Step 1: no irredundant broadcast of H' weights both twins. Suppose f is a broadcast with $f(p), f(p') \geq 1$; by the automorphism we may assume $f(p) \geq f(p')$. If $f(p') = 1$ then $\Omega_f(p') = N_f(p') = \{p', p, q\} \subseteq N_f(p)$, because $f(p) \geq 1$ and $d(p, p') = d(p, q) = 1$. If $f(p') \geq 2$ then $p \notin B_f(p')$, since $d(p', p) = 1 < f(p')$; and every $z \in B_f(p')$ satisfies $z \notin \{p, p'\}$ and so, by (1), $d(p, z) = d(p', z) = f(p') \leq f(p)$, i.e. $z \in N_f(p)$. Either way $\Omega_f(p') \subseteq N_f(p)$ with $p \in V_f^+ \setminus \{p'\}$, hence $\text{PB}_f(p') = \emptyset$ and f is redundant.

Step 2: $\Gamma_b(H') \geq \Gamma_b(H)$. Let f be a minimal dominating broadcast of H of weight $\Gamma_b(H)$ and extend it by $f(p') = 0$. It is a legal broadcast of H' because eccentricities did not change. It dominates p' : some $z \in V_f^+$ has $d_H(z, p) \leq f(z)$, and then $d(z, p') = d(z, p) \leq f(z)$ if $z \neq p$, while $d(p, p') = 1 \leq f(p)$ if $z = p$. It is still irredundant: for $w \in V_f^+$ the sets Ω and N_f computed in H' meet $V(H)$ in the sets computed in H , so any $u \in \text{PB}_f^H(w)$ still lies in $\text{PB}_f^{H'}(w)$.

Step 3: $\Gamma_b(H') \leq \Gamma_b(H)$. Let g be a minimal dominating broadcast of H' of weight $\Gamma_b(H')$. By Step 1 and the automorphism we may assume $g(p') = 0$. We claim

$$(2) \quad p' \in \text{PB}_g(w) \implies p \in \text{PB}_g(w) \quad \text{for every } w \in V_g^+.$$

Assume $p' \in \text{PB}_g(w)$. Then $d(z, p') > g(z)$ for every $z \in V_g^+ \setminus \{w\}$, hence by (1) also $d(z, p) > g(z)$ for every such z other than p ; and $p \neq w$ would force $g(p) = 0$, since $g(p) \geq 1$ would put $p' \in N_g(p)$ with $p \in V_g^+ \setminus \{w\}$. So $p \notin N_g(z)$ for all $z \in V_g^+ \setminus \{w\}$ in either case. It remains to see $p \in \Omega_g(w)$. If $w = p$ then $g(p) \geq 1$; here $g(p) \geq 2$ is impossible, because $\Omega_g(p) = B_g(p)$ would then exclude p' at distance 1, so $g(p) = 1$ and $p \in N_g(p) = \Omega_g(p)$. If $w \neq p$ and $g(w) \geq 2$ then $d(w, p') = g(w)$, so $d(w, p) = g(w)$ by (1)

and $p \in B_g(w) = \Omega_g(w)$. If $w \neq p$ and $g(w) = 1$ then $p' \in N_g(w)$ forces $w \in \{p, q\}$, hence $w = q$ and $p \in N_g(q) = \Omega_g(w)$. This proves (2).

Now restrict g to $V(H)$. The restriction is a legal broadcast of H of the same weight, and it dominates H because $p' \notin V_g^+$ and distances did not change. For $w \in V_g^+$ the set $\text{PB}_g(w)$ is non-empty; if it meets $V(H)$ we are done, and if $\text{PB}_g(w) = \{p'\}$ then (2) gives $p \in \text{PB}_g(w)$, a contradiction. So $\text{PB}_g(w) \cap V(H) \neq \emptyset$, which is exactly the private boundary of the restriction computed in H . The restriction is therefore a minimal dominating broadcast of H of weight $\Gamma_b(H')$. $\square$

Proof of $\Gamma_b(G') = 5$, and of Theorem 1. $\Gamma_b(G') \geq 5$ is witnessed by $g(v) = g(x_1) = g(x_2) = g(w_1) = g(w_2) = 1$, which dominates G' and has $\text{PB}_g(v) = \{v, u_1, u_2\}$, $\text{PB}_g(x_i) = \{x_i, x'_i, y_i\}$ and $\text{PB}_g(w_i) = \{w_i, w'_i, z_i\}$, all non-empty.

For $\Gamma_b(G') \leq 5$, apply Lemma 2 four times, once for each of the end-vertices x_1, x_2, w_1, w_2 of G_1 — each doubling is legitimate because the vertex doubled is still an end-vertex of the graph reached so far, the four being pairwise non-adjacent — to get $\Gamma_b(G') = \Gamma_b(G_1) = 5$, the last equality being [2, Proposition 4.8] at $k = 1$. Independently, $\Gamma_b(G') = 5$ is returned by the census of Section 4.

Combining with $\alpha_{\text{bnr}}(G') \geq 6$ gives $\alpha_{\text{bnr}}(G') \geq 6 > 5 = \Gamma_b(G')$ while $\delta(G') = 2$, which answers that question negatively. The census also gives the exact values $\alpha_{\text{bnr}}(G') = 6$ and $\alpha_{\text{bn}}(G') = 7$. $\square$

Remark. G_1 itself already has $\alpha_{\text{bnr}}(G_1) = 6 > 5 = \Gamma_b(G_1)$ — both values are printed in [2] — but $\delta(G_1) = 1$, which is precisely the case [2] excludes and the reason the question is asked. The doubling raises the minimum degree to 2 while preserving all distances among the vertices of G_1 , so that the source's own broadcast survives verbatim.

Remark. Only the second sentence of [2, Question 4], in the repaired reading of Section 1, is treated here. The first, for 2-connected G , is not addressed: G' has four cut vertices, and $\delta \geq 2$ is strictly weaker than 2-connectedness. No minimality is claimed. The definitions of bn-independence, irredundance, α_{bnr} and Γ_b used above, including the convention that a broadcaster of strength 1 lies in its own private boundary and the characterisation of a minimal dominating broadcast as a dominating irredundant one, are transcribed from [2, 1] and not re-derived.

4. VERIFICATION

Every quantity printed above for G' and for G_1 is re-derived from the 26-edge list of Section 2 by the program `verify.py` accompanying this note, whose recorded run is `verify.output.txt`. The program is Python 3.9+, standard library only, exact integer and bitmask arithmetic, no external data file; it runs in about 16 seconds and reports 53 checks, all passing, exiting 0 only if every check passes. It

- rebuilds G' from the printed edge list and confirms that the printed graph6 string decodes to the same *labelled* graph; re-derives the degree and eccentricity tables, $\delta = 2$, $\text{diam} = 4$, the cut-vertex set $\{y_1, y_2, z_1, z_2\}$ and the count 1,474,560,000;
- confirms the doubling structure: deleting the four primed vertices leaves 11 vertices and 18 edges with end-vertex set exactly $\{x_1, x_2, w_1, w_2\}$, each primed vertex is a true twin, and no distance and no eccentricity inside $V(G_1)$ changes;
- re-derives every set printed in the proof of Theorem 1 — N_f, B_f, PB_f for all four broadcasters, the six pairwise distances, the weight, the irredundance, the bn-independence, and that v is the unique unheard vertex — and the private boundaries of the weight-5 dominating broadcast;
- runs its own census of the broadcast space and returns $\Gamma_b(G') = 5$, $\alpha_{\text{bnr}}(G') = 6$, $\alpha_{\text{bn}}(G') = 7$ with 283 minimal dominating, 1140 bn-independent irredundant and 1416 bn-independent broadcasts, and $\Gamma_b(G_1) = 5$, $\alpha_{\text{bnr}}(G_1) = 6$, $\delta(G_1) = 1$; it also verifies over all 1780 irredundant broadcasts of G' that none weights both members of a twin pair, which is Step 1 of Lemma 2 at this graph;
- validates the census enumerator. The census prunes on irredundance and on bn-independence, both monotone conditions, so no member of either family is missed; to check the implementation rather than the argument, the same quantities are recomputed for G_1 by an *unpruned* walk of all 3,686,400 of its broadcasts and agree on all seven, and the enumerator reproduces published values for P_6, P_7, C_6 (cf. [4]) and the 3×3 and 3×4 grids. No unpruned walk of G' itself is performed;
- checks $\Gamma_b(H + p') = \Gamma_b(H)$ for every connected labelled graph H on $3, \dots, 6$ vertices and every end-vertex p of H , with no mismatch. This corroborates Lemma 2; the lemma itself rests on its proof.

The program closes by printing its own scope.

REFERENCES

- [1] D. Erwin, *Dominating broadcasts in graphs*, Bull. Inst. Combin. Appl. **42** (2004), 89–105.
- [2] C. M. Mynhardt and L. Neilson, *Comparing upper broadcast domination and boundary independence broadcast numbers of graphs*, Trans. Combin. **13** (2024), no. 1, 105–126. doi:10.22108/toc.2023.127904.1836; e-print arXiv:2104.02257v2, from which the statement numbers and the verbatim text of Question 4 are quoted.
- [3] L. Neilson, Ph.D. thesis, University of Victoria, 2019; cited in [2] as the source of the phenomenon $\alpha_{\text{bnr}} > \Gamma_b$.
- [4] S. Bouchouika, I. Bouchemakh, and É. Sopena, *Broadcasts on paths and cycles*, Discrete Appl. Math. **283** (2020), 375–395.